

Erdos Solver Pipeline

Browser automation that submits open Erdos problems to ChatGPT and DeepSeek, extracts the answers, saves them, and labels each chat [solved]/[unsolved] + confidence.

Overview

The pipeline runs in two stages per platform. A submit stage opens one chat per problem and sends the research-mode prompt. A rename stage revisits each chat, waits for generation to finish, extracts the answer, writes it to disk, and renames the chat with the verdict and confidence. ChatGPT and DeepSeek share the prompt and parsing logic via the common modules.

Components

- **erdos_common.py** Shared prompt, answer parsing (is_solved, extract_confidence), browser launch, and ChatGPT page helpers.
- **deepseek_common.py** DeepSeek page helpers; re-uses the prompt and parsers from erdos_common.
- **solve_submit.py / deepseek_submit.py** Submit problems to chats (no waiting). Record each problem -> chat URL in a chat map.
- **solve_rename.py / deepseek_rename.py** Watch the chats, save answers, and rename with the result label.
- **fetch_erdos.py / fetch_categories.py** Download and categorize the Erdos problem set into erdos_problems/.

How to run

1. One-time login (opens a browser to authenticate):

```
python3 solve_submit.py --login
python3 deepseek_submit.py --login
```

2. Submit a batch of open problems (reverse order shown):

```
python3 solve_submit.py --reverse --start 0 --limit 208
python3 deepseek_submit.py --reverse --start 0 --limit 208 --delay 60
```

3. Collect answers and rename chats (run as a watch loop):

```
python3 solve_rename.py --watch --interval 60
python3 deepseek_rename.py --watch --interval 45
```

Output

Solutions are written to erdos_problems/solutions/<category>/ (ChatGPT) and erdos_problems/solutions_deepseek/<category>/ (DeepSeek). Each chat title becomes:

```
Erdos #<n> [solved] 88%
Erdos #<n> [unsolved] 12%
```

Configuration & privacy

The private ChatGPT project URL is read from the CHATGPT_PROJECT_URL environment variable (it is not committed). Browser profiles, chat maps, and logs are git-ignored because they contain session cookies and private conversation IDs.

```
export CHATGPT_PROJECT_URL="https://chatgpt.com/g/<your-project>/project"
```